

Burrowing Through The Network

*Contextualizing The Vulkan Leaks
& State-Sponsored Offensive Ops*



Joe Slowik

Threat Intelligence Manager



Today's Agenda

- 1 The Vulkan Files
- 2 Vulkan & Russian Cyber Operations
- 3 Orienting Vulkan Capabilities In Cyber History
- 4 Future Of Offensive Cyber Operations
- 5 Conclusions

01

The Vulkan Files



What Are The Vulkan Leaks?

The Vulkan Files





»Vulkan Files«

Stromnetze werden lahmgelegt, Satellitenverbindungen gestört und Rechner im Bundestag gehackt. Seit Jahren führt Russland einen Cyberkrieg gegen den Westen.

Koordiniert vom SPIEGEL hat ein Journalistenteam – darunter der »Guardian«, das ZDF, der österreichische »Standard«, die Schweizer Tamedia-Gruppe, die »Washington Post«, die »Süddeutsche Zeitung« und »Le Monde« – interne Unterlagen der Firma NTC Vulkan

What Are The Vulkan Leaks?

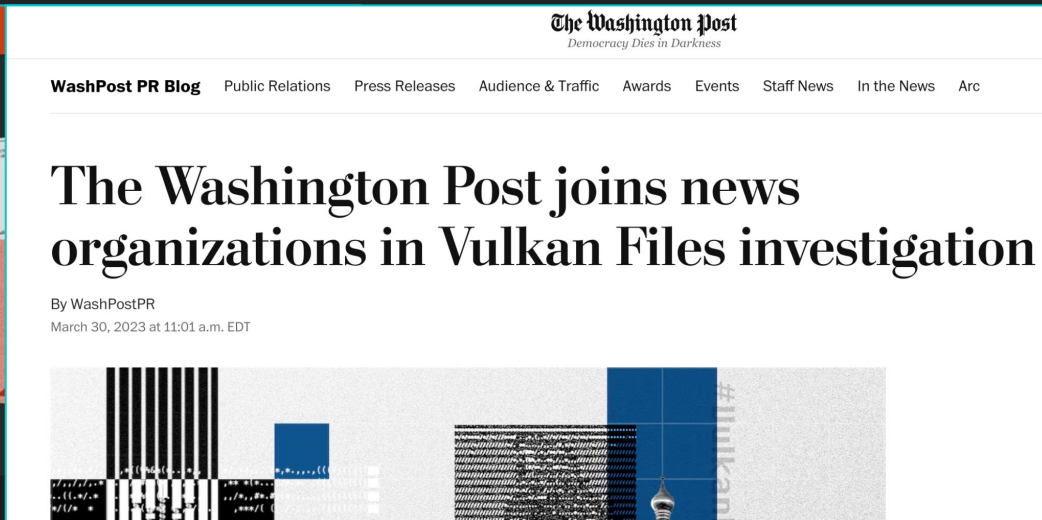
The Vulkan Files



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Platform Solutions Intelligence Services Resources

Contracts Identify Cyber Operations Projects from Russian Company NTC Vulkan

ALDEN WAHLSTROM, GABBY RONCONE, KEITH LUNDEN, DANIEL KAPELLMANN ZAFRA

MAR 30, 2023 | 9 MIN READ

#OPERATIONAL TECHNOLOGY #ICS #THREAT INTELLIGENCE #RUSSIA

As a part of Mandiant's research on Russian cyber and information operations (IO) capabilities, Mandiant worked with a collective of media outlets, including Papertrail Media, Der Spiegel, Le Monde, and Washington Post, to analyze several documents belonging to a Russian IT contractor named NTC Vulkan (Russian: HTLI Вулкан). The documents detail project requirements contracted with the Russian Ministry of Defense, including in at least one instance for GRU Unit 74455, also known as [Sandworm Team](#). These projects include tools, training programs, and a red team platform for exercising various types of offensive cyber operations,



Democracy Dies in Darkness

Intelligence & Traffic Awards Events Staff News In the News Arc

Post joins news outlet in Vulkan Files investigation



What Are The Vulkan Leaks?

The Vulkan Files

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NTC Vulkan

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The Vulkan Files

NTC Vulkan

The Leaks

What Are The Vulkan Leaks?

The Vulkan Files

NTC Vulkan

The Leaks

**The
Significance**

What Are The Vulkan Leaks?

The Vulkan Files

NTC Vulkan

Computer Technology Firm, Founded In 2010 By Russian Military Veterans With Approval To Perform Classified Work.

The Leaks

The Significance

What Are The Vulkan Leaks?

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The Leaks

Unknown Entity Leaked Project Information And Documentation For Various Projects To German Journalists

The Significance

What Are The Vulkan Leaks?

The Vulkan Files

NTC Vulkan

Computer Technology Firm, Founded In 2010 By Russian Military Veterans With Approval To Perform Classified Work.

The Leaks

Unknown Entity Leaked Project Information And Documentation For Various Projects To German Journalists

The Significance

Rarely Do We See The "Nuts & Bolts" Of CNO Tool Development & Procurement Processes - Lots Of Data To Go Through!

NTC Vulkan

The Vulkan Files



Programs Of Interest

The Vulkan Files

Programs Of Interest

The Vulkan Files

Scan-V

Programs Of Interest

The Vulkan Files

Scan-V

Amesit

Programs Of Interest

The Vulkan Files

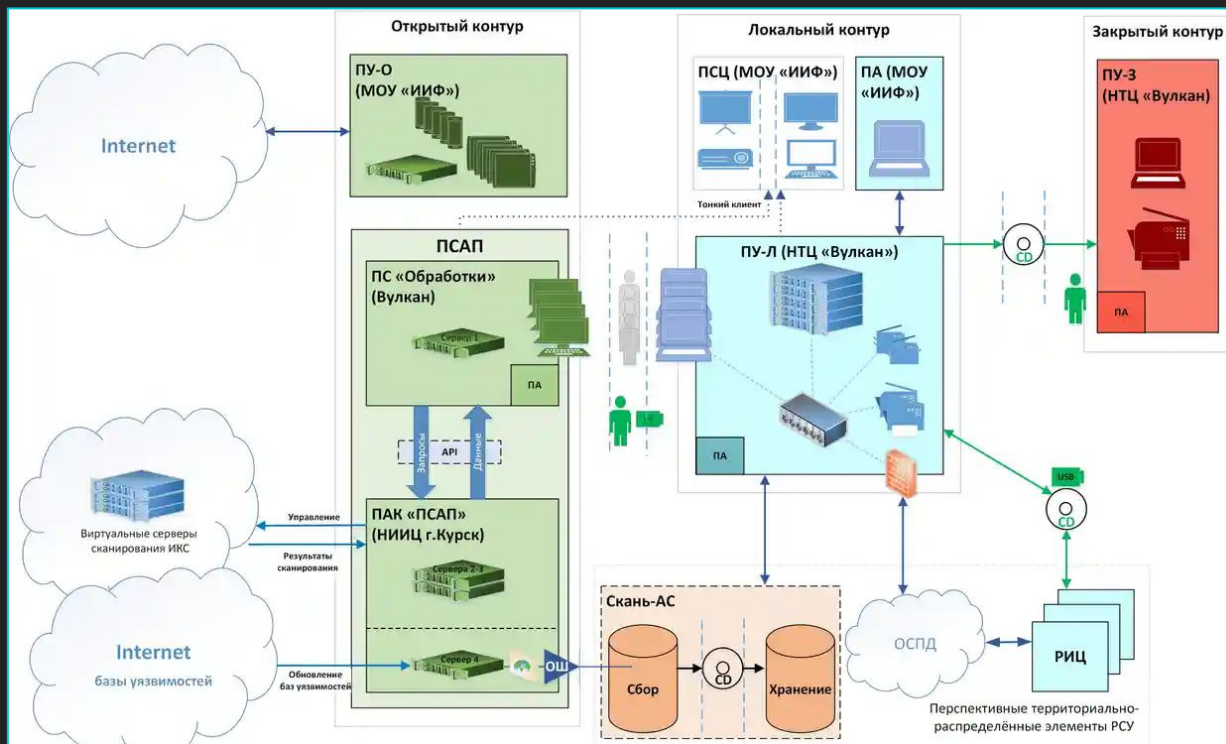
Scan-V

Amesit

Krystal-2V

Scan-V

The Vulkan Files



Scan-V

The Vulkan Files

Scan-V

The Vulkan Files

Design

Scan-V

The Vulkan Files

Design

- Distributed, Multi-Component System Crossing Various Information Boundaries
- Designed For Automated Tasking & Action Within Pre-Programmed Capabilities

Scan-V

The Vulkan Files

Design

- Distributed, Multi-Component System Crossing Various Information Boundaries
- Designed For Automated Tasking & Action Within Pre-Programmed Capabilities

Purpose

Scan-V

The Vulkan Files

Design

- Distributed, Multi-Component System Crossing Various Information Boundaries
- Designed For Automated Tasking & Action Within Pre-Programmed Capabilities

Purpose

- Combine External Scanning Functionality With Catalog Of Vulnerable Software & Exploits
- Automate Or Increase The Efficiency Of Cyber Operations, Infrastructure Harvesting

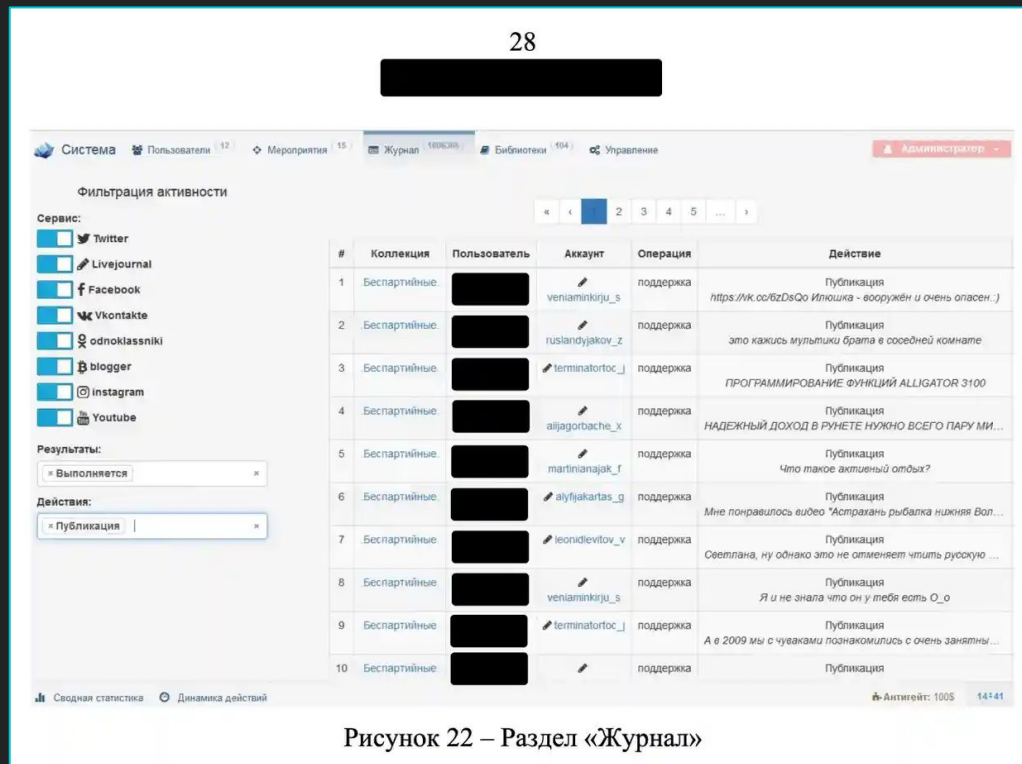


Рисунок 22 – Раздел «Журнал»

Amesit

The Vulkan Files

Amesit

The Vulkan Files

Information Operations & Collection Platform

Information Operations & Collection Platform



**Capable Of Capturing & Proxying Communication
Streams For Collection & Manipulation**

Information Operations & Collection Platform



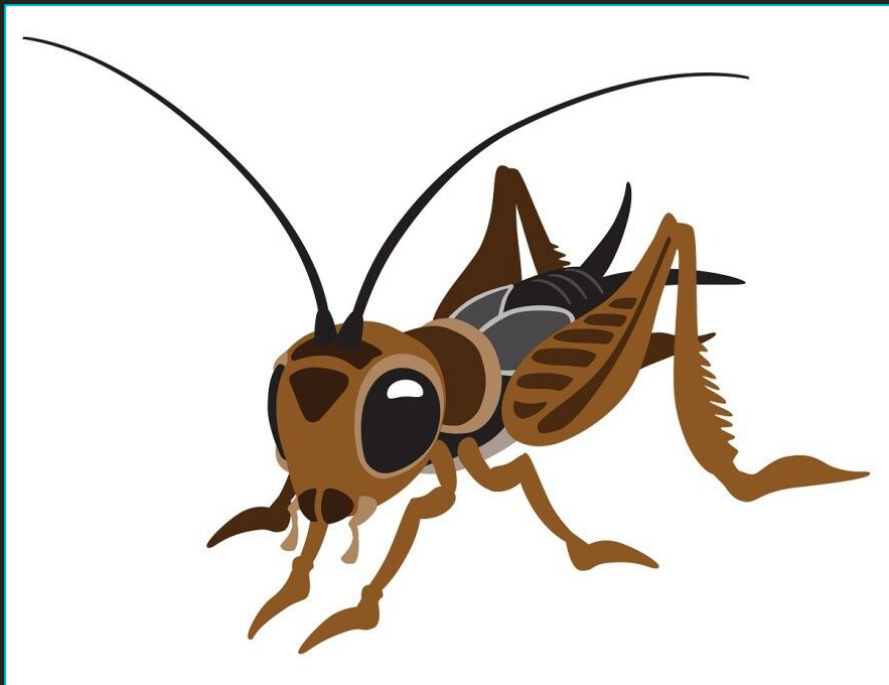
Capable Of Capturing & Proxying Communication Streams For Collection & Manipulation



Potential Applications In Both Foreign & Domestic Targeting

Response To The Vulkan Leaks

The Vulkan Files



Leaks Passed With Limited Interest???

The Vulkan Files

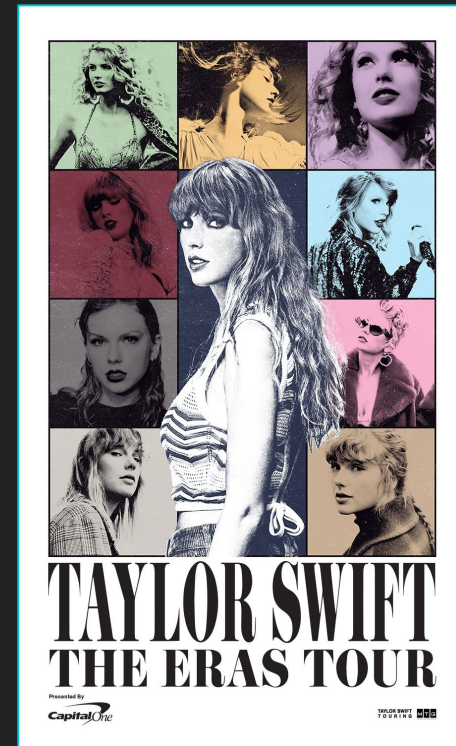
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The Vulkan Files



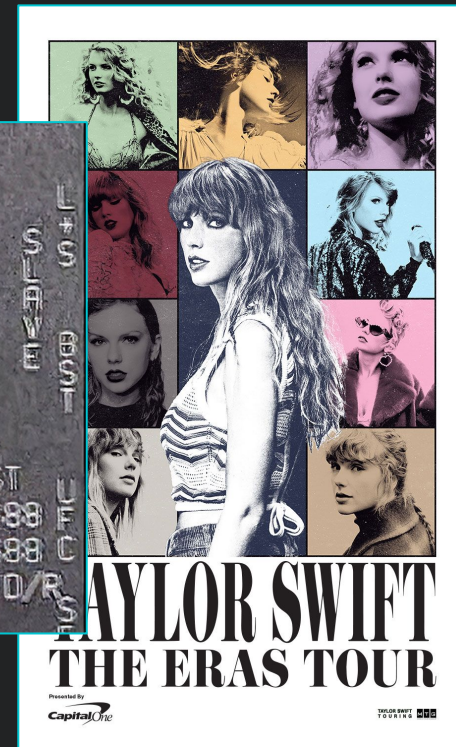
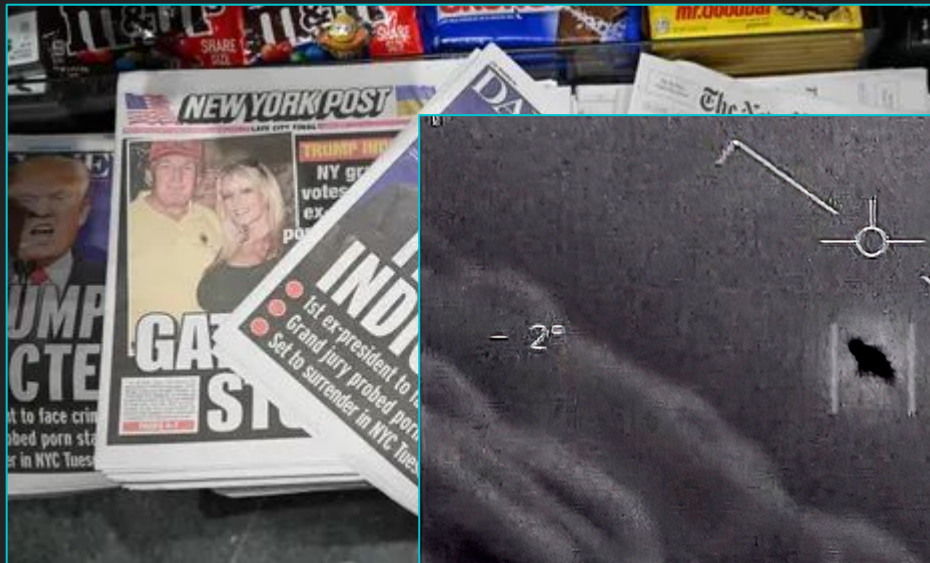
Leaks Passed With Limited Interest???

The Vulkan Files



Leaks Passed With Limited Interest???

The Vulkan Files



Diving Into The Vulkan Leaks

The Vulkan Files

The Vulkan Leaks Represent A Significant Event In The History Of Cyber Operations - We Need To Review Just What's Inside!

02

Vulkan & Russian Cyber Operations



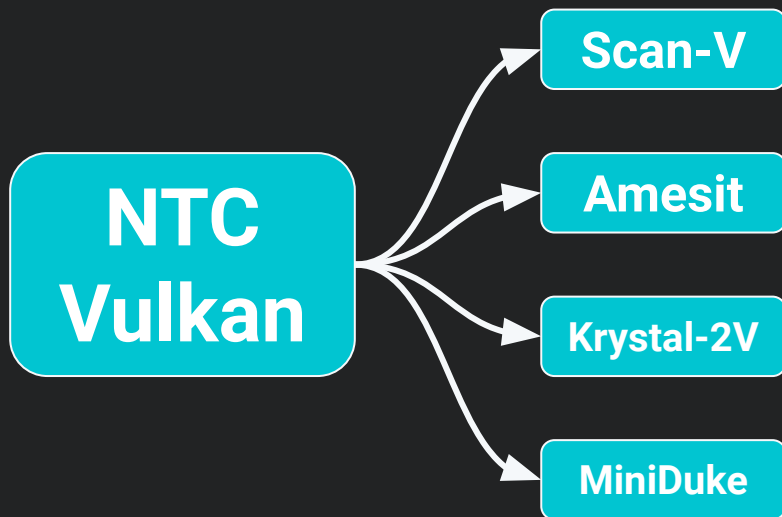
Diving Into The Vulkan Leaks

Vulkan & Russian Cyber Operations

**NTC
Vulkan**

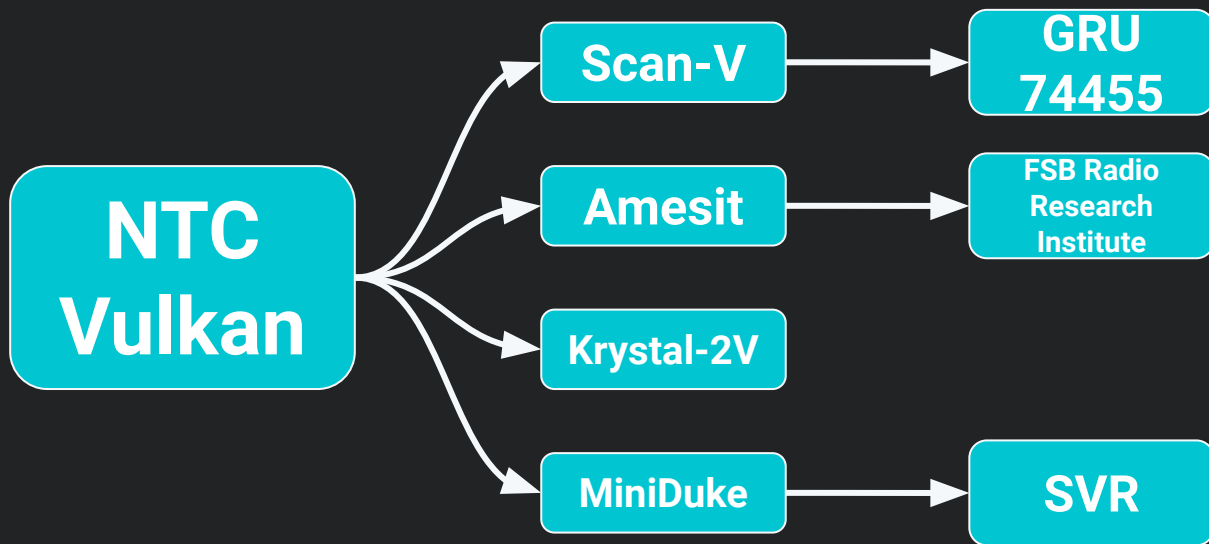
Diving Into The Vulkan Leaks

Vulkan & Russian Cyber Operations



Diving Into The Vulkan Leaks

Vulkan & Russian Cyber Operations



Vulkan & Russia's "Cyber Hydra"

Vulkan & Russian Cyber Operations

For Various Reasons, Research Institutes & External Entities *Generally* Perform Dedicated Work For One Element Of Russia's Cyber Forces...

Vulkan & Russia's "Cyber Hydra"

Vulkan & Russian Cyber Operations

For Various Reasons, Research Institutes & External Entities *Generally* Perform Dedicated Work For One Element Of Russia's Cyber Forces...

NTC Vulkan Worked With Each Element Of Russian External Cyberpower - GRU, FSB, & SVR - Not COMPLETELY Unheard Of, But Rare!

RU Cyber Ops & The Private Sector

Vulkan & Russian Cyber Operations

RU Cyber Ops & The Private Sector

Vulkan & Russian Cyber Operations

**TsNIIKhM &
XENOTIME**

RU Cyber Ops & The Private Sector

Vulkan & Russian Cyber Operations

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**ODT & FSB's
Fronton Botnet**

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Vulkan & Russian Cyber Operations

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FSB**

**INCONTROLLER /
PIPEDREAM?**

RU Cyber Ops & The Private Sector

Vulkan & Russian Cyber Operations

CyberWatchers @cyber_watchers · Jan 17

We tweeted in July about the development of a variant to the malware project Drovorub-A1 by Russian tech company AST (ACT).

CyberWatchers @cyber_watchers · Jul 14, 2022

If the Russian Intelligence Services work with other companies, which ones? According to the US, one company working with the FSB, GRU and SVR is Advanced System Technologies (AST).

[Show this thread](#)

SVA Institute & SVR

Kvant Institute &

Frontol

CyberWatchers @cyber_watchers · Jan 17

Over 3 years of development, the project has cost 54.7 million rubles (approx 900,000 USD). Given that AST also received 20 million rubles as part of the INCONTROLLER project development, it seems that AST has not provided much value for money to the GRU.

1 1 206

Privatization Of Cyber Capability Dev

Vulkan & Russian Cyber Operations

**NTC Vulkan Is Both An Example And A Pioneer Of
Increasingly Outsourced Cyber Development &
Engineering Operations For State-Directed
Operations!**

Privatization Of Cyber Capability Dev

Vulkan & Russian Cyber Operations

NTC Vulkan Is Both An Example And A Pioneer Of
Increasingly Out-Of-Country Development &
Engineering Operations!

**NOTE: This Is NOT
Just Russia!**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

**Sandworm
Historically
Leverages
Compromised,
Legitimate
Infrastructure**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

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**Multiple
Examples
From
Industroyer To
CyclopsBlink**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

**Sandworm
Historically
Leverages
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**Multiple
Examples
From
Industroyer To
CyclopsBlink**



**Operations
Can Scale
More
Effectively Via
Automation!**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

Stockpile Vulnerabilities

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

Stockpile Vulnerabilities



**Identify Vulnerable
Nodes**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations

Stockpile Vulnerabilities

```
graph TD; A[Stockpile Vulnerabilities] --> B[Identify Vulnerable Nodes]; B --> C[Marshall Victims Into Operational Proxy Chains];
```

**Identify Vulnerable
Nodes**

**Marshall Victims Into
Operational Proxy Chains**

Scan-V & Sandworm

Vulkan & Russian Cyber Operations



Scan-V & Sandworm

Vulkan & Russian Cyber Operations



The “So What” Behind Vulkan Items

Vulkan & Russian Cyber Operations

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Vulkan & Russian Cyber Operations

Focus On Scalability & Control Of Widespread Operations

The “So What” Behind Vulkan Items

Vulkan & Russian Cyber Operations

Focus On Scalability & Control Of Widespread Operations



Enables Greater Efficiency & Extent Of Cyber Activity

The “So What” Behind Vulkan Items

Vulkan & Russian Cyber Operations

Focus On Scalability & Control Of Widespread Operations



Enables Greater Efficiency & Extent Of Cyber Activity



Move Beyond “One Operator, One Op” To Massive Distributed Campaigns

03

—

Vulkan In Cyber History



Is Vulkan Activity Unique?

Vulkan In Cyber History

Is Vulkan Activity Unique?

Vulkan In Cyber History



Scan-V - Similar Examples?

Vulkan In Cyber History

Scan-V - Similar Examples?

Vulkan In Cyber History

Russia!

Scan-V - Similar Examples?

Vulkan In Cyber History

Russia!

China!

Scan-V - Similar Examples?

Vulkan In Cyber History

Russia!

China!

US???

Scan-V - Similar Examples?

Vulkan In Cyber History



Russia & Cyber Manipulation

Vulkan In Cyber History

Russia & Cyber Manipulation

Vulkan In Cyber History

**Building
Compromised
Networks**

Russia & Cyber Manipulation

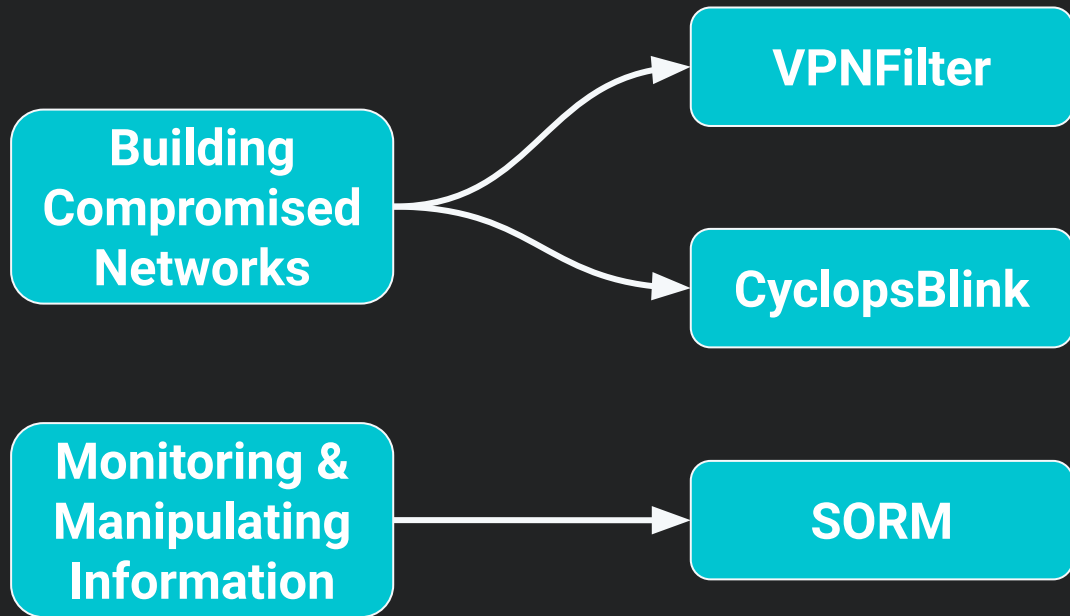
Vulkan In Cyber History

**Building
Compromised
Networks**

**Monitoring &
Manipulating
Information**

Russia & Cyber Manipulation

Vulkan In Cyber History



VPNFilter & CyclopsBlink

Vulkan In Cyber History

VPNFilter & CyclopsBlink

Vulkan In Cyber History

Network-Device Targeting Capabilities

VPNFilter & CyclopsBlink

Vulkan In Cyber History

Network-Device Targeting Capabilities



**Leverage Compromised Nodes To Proxy Traffic, Or
DDoS Activity**

VPNFilter & CyclopsBlink

Vulkan In Cyber History

Network-Device Targeting Capabilities



**Leverage Compromised Nodes To Proxy Traffic, Or
DDoS Activity**



**Evidence Of Potential Further Capabilities
(VPNFilter Modbus Capture)**

VPNFilter & CyclopsBlink To Scan-V

Vulkan In Cyber History

**Scan-V Represents A Mechanism To Automate,
Control, & Manage A Network Of Collection Nodes &
ORBs For Cyber Operations - Scalability &
Efficiency!**

SORM

Vulkan In Cyber History

ANDREI SOLODOV

IRINA BOROGAN

SECURITY NOV 1, 2012 6:38 AM

The Kremlin's New Internet Surveillance Plan Goes Live Today

On the surface, it's all about protecting Russian kids from internet pedophiles. In reality, the Kremlin's new "Single Register" of banned websites, which goes into effect Nov. 1, will wind up blocking all kinds of online political speech. And, thanks to the spread of new internet-monitoring technologies, the Register could well become a tool for spying on millions of Russians.

ANDREI SOLODOV

IRINA BOROGAN

SECURITY NOV 1 2012 6:38 AM

The Kremlin's

On the surface, it's all about pro effect Nov. 1, will wind up block well become a tool for spying on

LIBRARY

LIBRARY OF CONGRESS

Everything



Share

Listen to this page

ARTICLE

ECHR, Russian Federation: Breaches of Human Rights in Surveillance Legislation

(Mar. 2, 2016) The European Court of Human Rights (ECHR), in a decision issued on December 4, 2015, in the case of *Roman Zakharov v. Russia*, ruled on the legality of Russia's regulations on administering the System for Ensuring Investigative Activities (SORM legislation) under article 34 of the Convention for the Protection of Human Rights and Fundamental Freedoms. ([Case of Roman Zakharov v. Russia, App. No. 47143/06](#) (Eur. Ct. H. Rts., Dec. 4, 2015), HUDOC; [Convention for the Protection of Human Rights and Fundamental Freedoms](#) (1950, as amended to June 1, 2010), ECHR website.) The ECHR held that the legislation "institutes a system which cannot protect individuals from secret surveillance" and "any person using mobile telephone services of Russian providers can have his or her mobile telephone communications intercepted, without ever being notified of the surveillance." (Case of Roman Zakharov v. Russia, ¶ 175.)

SORM To Amesit

Vulkan In Cyber History

Amesit Enables A Mobile, Flexible Deployment Of SORM-Like Capabilities, Including For Information & Influence Operation Purposes - Can Be “Forward Deployed!”

What About China?

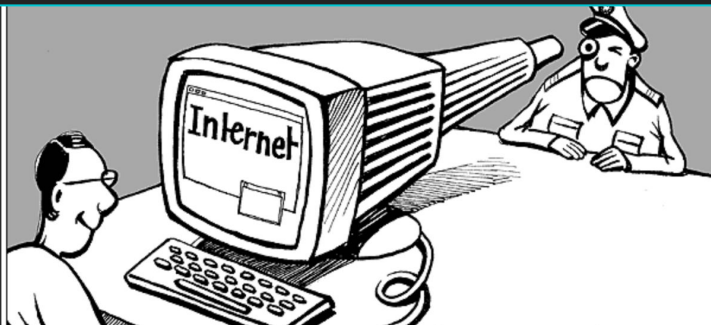
Vulkan In Cyber History

What About China?

Vulkan In Cyber History

Free speech vs Maintaining Social Cohesion

A Closer Look at Different Policies

[HOME](#)[EUROPEAN UNION POLICY](#)[CHINA'S GREAT FIREWALL](#)[US POLICY](#)[GOOGLE](#)

China's Great Firewall

- [Background Information](#)
- [Opinions](#)
- [References](#)

Background Information

China is known for its strict policies regarding information control in comparison to the regulations adopted in other countries. The Golden Shield Project, often called the "great firewall of China", is an initiative managed by the Ministry of Public Security division of the Chinese government. As the nickname implies, the focus of this project is to monitor and censor what can and cannot be seen through an online network in China. This project started in 1998 and is still continually improving in restriction techniques through multiple methods. The OpenNet Initiative performed an empirical study that concluded that China has "the most sophisticated content-filtering Internet regime in the world". Some technical methods used are IP blocking, which denies

What About China?

Vulkan In Cyber History

Free speech vs Maintaining Social Cohesion

A Closer Look at Different Policies



HOME

EUROPEAN UNION POLICY

CHINA'S GREAT FIREWALL

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munkschool
OF GLOBAL AFFAIRS & PUBLIC POLICY



UNIVERSITY OF
TORONTO

RESEARCH NEWS ABOUT



Research > Free Expression Online

China's Great Cannon

By Bill Marczak^{1, 2, 3}, Nicholas Weaver^{2, 3}, Jakub Dalek¹, Roya Ensafi⁴, David Fifield³, Sarah McKune¹, Arn Rey, John Scott-Railton¹, Ron Deibert¹, and Vern Paxson^{2, 3}

[1] Citizen Lab, Munk School of Global Affairs, University of Toronto [2] International Computer Science Institute

[3] University of California, Berkeley [4] Princeton University

April 10, 2015

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This post describes our analysis of China's "Great Cannon," our term for an attack tool that we identify as separate from, but co-located with, the Great Firewall of China. The first known usage of the Great Cannon is in the recent large-scale novel DDoS attack on both GitHub and servers used by GreatFire.org.

China's CNO & IO Operations

Vulkan In Cyber History

China's CNO & IO Operations

Vulkan In Cyber History

**Great Cannon
Is Nearly 10
Years Old Now
(Disclosed In
2015)**

China's CNO & IO Operations

Vulkan In Cyber History

**Great Cannon
Is Nearly 10
Years Old Now
(Disclosed In
2015)**



**Presumably
PRC Continues
To Invest &
Innovate In
Cyber
Operations**

China's CNO & IO Operations

Vulkan In Cyber History



China's CNO & IO Operations

Vulkan In Cyber History

**Great Firewall & Cannon Functionality Give Glimpses
Of Similar Ambitions & Programs - Main Question Is
Scalability & Management Of These Capabilities,
And Widespread Intrusion Campaigns**

The United States & FVEY

Vulkan In Cyber History

**Hey - Remember The Snowden Leaks
(Among Other Items)?**

The United States & FVEY

Vulkan In Cyber History

Hey - Remember Vulkan Leaks

**NOTE THAT THE FOLLOWING SLIDES ARE
BASED ON PUBLIC COMMENTARY AND
POSTING HERE NEITHER CONFIRMS NOR
DENIES THE ACCURACY OR VERACITY OF
THESE ITEMS!**

NSA Leaks & Implications

Vulkan In Cyber History

NSA Leaks & Implications

Vulkan In Cyber History

Popular Conception

NSA Leaks & Implications

Vulkan In Cyber History

Popular Conception

- Emphasis On Privacy Violations & Other Elements Detrimental To Open Democracies
- This Is Certainly Important - But Hardly The Limit Of What Was Disclosed

NSA Leaks & Implications

Vulkan In Cyber History

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Additional Details

NSA Leaks & Implications

Vulkan In Cyber History

Popular Conception

- Emphasis On Privacy Violations & Other Elements Detrimental To Open Democracies
- This Is Certainly Important - But Hardly The Limit Of What Was Disclosed

Additional Details

- Snowden (And Potentially Others) Disclosed Significant Information On CNO Capabilities
- Less-Heralded But Arguably More Significant Were Widespread, Automated Exploitation & Control Frameworks

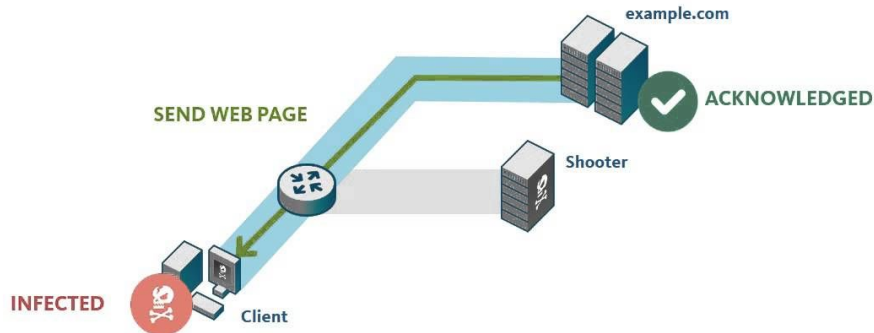
A QUANTUM Of Exploitation

Vulkan In Cyber History

A QUANTUM Of Exploitation

Vulkan In Cyber History

How a Quantum Insert works



Malicious response is treated as actual response



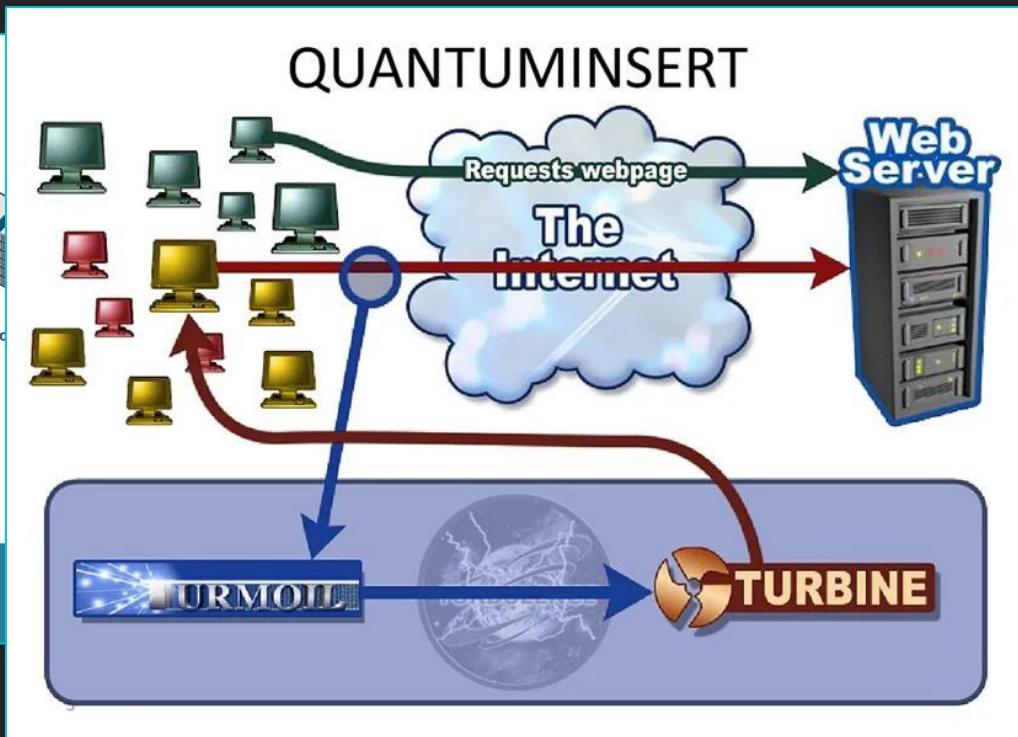
A QUANTUM Of Exploitation

Vulkan In Cyber History

How a Quantum Insert works



Malicious response is treated as actual response



Automated Exploitation At Scale

Vulkan In Cyber History

Automated Exploitation At Scale

Vulkan In Cyber History

**QUANTUM, TURBINE, Etc. Represent
Automated Exploitation Systems**

Automated Exploitation At Scale

Vulkan In Cyber History

**QUANTUM, TURBINE, Etc. Represent
Automated Exploitation Systems**



**Disclosed Publicly In Early 2010's,
Possibility Adversaries Identified
Earlier**

Automated Exploitation At Scale

Vulkan In Cyber History

**QUANTUM, TURBINE, Etc. Represent
Automated Exploitation Systems**



**Disclosed Publicly In Early 2010's,
Possibility Adversaries Identified
Earlier**



**Alleged FVEY "State Of The Art" Now
Reflected In CN, RU Operations**

Snowden, Vulkan, & Beyond

Vulkan In Cyber History

**IF TRUE - The Snowden Leaks Were Arguably A
Disaster For US, Related CNO & SIGINT Capabilities
By Disclosing Methodology...**

Snowden, Vulkan, & Beyond

Vulkan In Cyber History

IF TRUE - The Snowden Leaks Were Arguably A Disaster For US, Related CNO & SIGINT Capabilities By Disclosing Methodology...

...But It Also Appears That Multiple Parties Were Taking Notes On How To Build Similar (Or Even More Ambitious) Programs Based On Disclosures

04

The Future Of Offensive Cyber

Popular Conceptions Of CNO

The Future Of Offensive Cyber



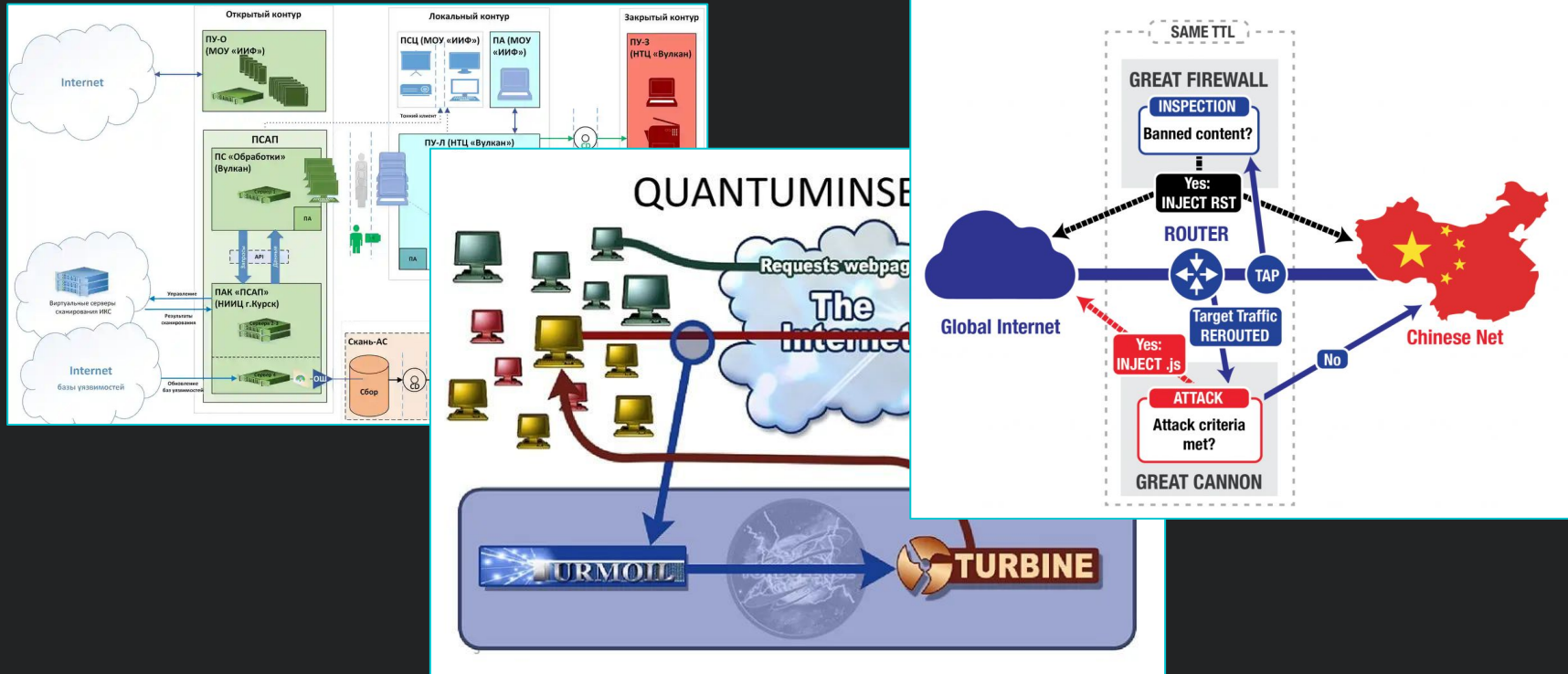
Realistic Depictions Of CNO

The Future Of Offensive Cyber



Future Applications Of CNO

The Future Of Offensive Cyber



The Road To Scalable Cyber

The Future Of Offensive Cyber

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

Targeting

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

Targeting

**Operational
Security**

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

Cyber Talent Is Expensive - Codifying Capabilities In Technology & Programs Is Key To Expansive & Efficient Action!

Targeting

Operational Security

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

Cyber Talent Is Expensive - Codifying Capabilities In Technology & Programs Is Key To Expansive & Efficient Action!

Targeting

Global Ambitions Require Global Actions. Assumptions On Targeting & Focus Go Out The Window, Especially When Targets Can Be "Means To An End!"

Operational Security

The Road To Scalable Cyber

The Future Of Offensive Cyber

Personnel

Cyber Talent Is Expensive - Codifying Capabilities In Technology & Programs Is Key To Expansive & Efficient Action!

Targeting

Global Ambitions Require Global Actions. Assumptions On Targeting & Focus Go Out The Window, Especially When Targets Can Be "Means To An End!"

Operational Security

Top-Tier Actors Know How Defenders & CTI Analysts Operate - They Will Work To Evade Known Analytical Tradecraft & Minimize "Touchpoints" Wherever Possible!

NTC Vulkan & CNO

The Future Of Offensive Cyber

Vulkan Capabilities Are *LAGGING* Indicators - This Activity Has Already Taken Place!

NTC Vulkan & CNO

The Future Of Offensive Cyber

Vulkan Capabilities Are *LAGGING* Indicators - This Activity Has Already Taken Place!

We Need To Anticipate Greater Degrees Of Automation, Queueing, & Reactive Targeting!

What Should We Expect?

The Future Of Offensive Cyber

What Should We Expect?

The Future Of Offensive Cyber

**Greater Automation
& Improved
Scalability**

What Should We Expect?

The Future Of Offensive Cyber

**Greater Automation
& Improved
Scalability**

**Marshalling The
“Neutral Web” For
Offensive Action**

What Should We Expect?

The Future Of Offensive Cyber

**Greater Automation
& Improved
Scalability**

**Marshalling The
“Neutral Web” For
Offensive Action**

**Reduced Reliance
On “Rockstars” &
Commoditization**

What Should We Expect?

The Future Of Offensive Cyber

**Greater Automation
& Improved
Scalability**

**Marshalling The
“Neutral Web” For
Offensive Action**

**Reduced Reliance
On “Rockstars” &
Commoditization**

**Non-Bullshit
Applications For AI**

05

Conclusions

Where Are We Now?

Conclusions



Where Are We Now?

Conclusions

But...

Capability Inspiration & Proliferation

Conclusions

Capability Inspiration & Proliferation

Conclusions

Adversaries Learn From Operations & Open Source Research

Capability Inspiration & Proliferation

Conclusions

Adversaries Learn From Operations & Open Source Research



Today's Leak Or Disclosure Is Inspiration For Tomorrow's Capability & Evolution

Capability Inspiration & Proliferation

Conclusions

Adversaries Learn From Operations & Open Source Research



Today's Leak Or Disclosure Is Inspiration For Tomorrow's Capability & Evolution



CNO Is NOT A Static Field, But A Constantly Evolving One With Multiple Factors

Guidance For Defenders & Decision-Makers

Conclusions

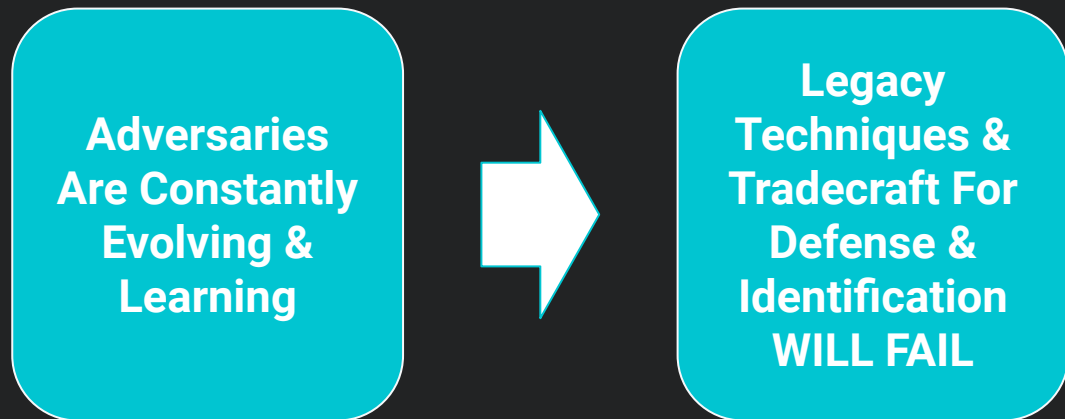
Guidance For Defenders & Decision-Makers

Conclusions

**Adversaries
Are Constantly
Evolving &
Learning**

Guidance For Defenders & Decision-Makers

Conclusions



Guidance For Defenders & Decision-Makers

Conclusions



Advice For Hackers & Red Teamers

Conclusions

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**Human-Driven Operations Are Rapidly
Being Replaced In High-Level Ops**

Advice For Hackers & Red Teamers

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**Human-Driven Operations Are Rapidly
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**Adversary Emulation Requires
Understanding Shifts In Threat Space**

Advice For Hackers & Red Teamers

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**Human-Driven Operations Are Rapidly
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**Adversary Emulation Requires
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**Testing & Probing Will Become More
Challenging!**

Where Do We Go From Here?

Conclusions

Networks Of Various Types Will Continue To Be Weaponized By Diverse Parties Either As End-Targets Or Means To Reaching Them...

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Networks Of Various Types Will Continue To Be Weaponized By Diverse Parties Either As End-Targets Or Means To Reaching Them...

...The Increasing Scale & Velocity Of Campaigns Will Make Defense & Response Challenging - But Not Impossible!

Questions?

Selected Resources

This Is Not A Complete List!

- The Vulkan Files - Papertrail Media (<https://www.papertrailmedia.de/investigations/vulkan-files/>)
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- "'Vulkan Files' Leak Reveals Putin's Global And Domestic Cyberwarfare Tactics" - Luke Harding, Stiliyana Simeonova, Manisha Ganguly, & Dan Sabbagh, The Guardian (<https://www.theguardian.com/technology/2023/mar/30/vulkan-files-leak-reveals-putins-global-and-domestic-cyberwarfare-tactics>)
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- CyberWatchers Twitter Thread (<https://threadreaderapp.com/thread/1615358254409986053.html>)
- "New VPNFilter Malware Targets At Least 500k Networking Devices Worldwide" - William Largent, Cisco Talos (<https://blog.talosintelligence.com/vpnfilter/>)
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- "China's Great Cannon" - Bill Marczak, Nicholas Weaver, Jakub Dalek, Roya Ensafi, David Fifield, Sarah McKune, Arn Rey, John Scott-Railton, Ron Deibert, & Vern Paxson, The Citizen Lab (<https://citizenlab.ca/2015/04/chinas-great-cannon/>)
- "A Close Look At The NSA's Most Powerful Internet Attack Tool" - Nicolas Weaver, Wired (<https://www.wired.com/2014/03/quantum/>)
- "NSA's Automated Hacking Engine Offers Hands-Free Pwning Of The World" - Sean Gallagher, Ars Technica (<https://arstechnica.com/information-technology/2014/03/nsas-automated-hacking-engine-offers-hands-free-pwning-of-the-world/>)
- "Hack Like The NSA: The Quantum Insert" - otw, HackersArise (<https://www.hackers-arise.com/post/2016/09/09/hack-like-the-nsa-the-quantum-insert>)